

```

1 namespace CSharp;
2 public class LinqObj
3 {
4     public static bool IsTeenAger(LinqObj.Student stud)
5     {
6         return stud.Age > 12 && stud.Age < 20;
7     }
8
9     public class Student
10    {
11        public int StudentID { get; set; }
12        public string StudentName { get; set; }
13        public int Age { get; set; }
14    }
15
16    public static void Main()
17    {
18        // Student collection
19        IList<Student> studentList = new List<Student>() {
20            new Student() { StudentID = 1, StudentName = "Dipak", Age = 13 } ,
21            new Student() { StudentID = 2, StudentName = "Rahul", Age = 20 } ,
22            new Student() { StudentID = 3, StudentName = "Jinal", Age = 18 } ,
23            new Student() { StudentID = 4, StudentName = "Dhruti", Age = 20 } ,
24            new Student() { StudentID = 5, StudentName = "Mayank", Age = 15 } ,
25            new Student() { StudentID = 6, StudentName = "Dinky", Age = 15 } ,
26            new Student() { StudentID = 7, StudentName = "Bony", Age = 12 } ,
27            new Student() { StudentID = 8, StudentName = "Sunil", Age = 15 }
28        };
29
30        // var filteredResult = from s in studentList select s;
31        //
32        // var filteredResult = from s in studentList
33        //                         where s.Age > 12 && s.Age <= 18
34        //                         select s;
35        //
36        // Func<Student, bool> isTeenAger = delegate (Student s)
37        // {
38        //     return s.Age > 12 && s.Age < 19;
39        // };
40        //
41        // var filteredResult = from s in studentList
42        //                         where isTeenAger(s)
43        //                         select s;
44        //
45        // var filteredResult = from s in studentList
46        //                         where IsTeenAger(s)
47        //                         select s;
48        //
49        // var filteredResult = studentList.Where(s => s.Age > 12 && s.Age < 20);
50        //
51        // var filteredResult = from s in studentList
52        //                         orderby s.Age descending, s.StudentName
53        //                         select s;
54        // Console.WriteLine("Teen age Students:");
55        //
56        // foreach (Student std in filteredResult)
57        // {
58        //     Console.WriteLine("ID :" + std.StudentID + " Name :" + std.StudentName + " Age :" + std.Age);
59        // }
60        //
61        // var std = (from s in studentList
62        //              where s.Age >12 && s.Age < 20
63        //              select s).FirstOrDefault();
64        // var std = (from s in studentList
65        //              where s.Age > 12 && s.Age < 20
66        //              select s).LastOrDefault();
67        // var std = (from s in studentList
68        //              where s.Age == 15
69        //              select s).SingleOrDefault();
70        // var filteredResult = studentList.OrderByDescending(s => s.Age).ThenBy(s => s.StudentName);
71        var groupedResult = from s in studentList group s by s.Age;
72        foreach (var ageGroup in groupedResult)
73        {
74            Console.WriteLine("Age Group: {0}", ageGroup.Key); //Each group has a key
75            Console.WriteLine(ageGroup.Max(s => s.StudentID));
76            Console.WriteLine(ageGroup.Count());
77
78            foreach (Student std in ageGroup) // Each group has inner collection
79                Console.WriteLine("ID :" + std.StudentID + "Name :" + std.StudentName + " Age :" + std.Age);
80        }
81        // Console.WriteLine(studentList.Sum(s => s.Age));
82        // Console.WriteLine(studentList.Max(s => s.Age));
83        // Console.WriteLine(studentList.Min(s => s.Age));
84        // Console.WriteLine(studentList.Average(s => s.Age));
85    }
86 }

```